

ANDREW P. FREIBURGER

Computational biologist building metabolic-modeling platforms for microbiome therapeutics

PhD Candidate, Chemical & Biological Engineering · Northwestern University

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EDUCATION

Northwestern University — Ph.D., Chemical and Biological Engineering · Tyo Lab	Sep 2023 – Present
University of Victoria — M.A.Sc., Civil Engineering · GPA 8.0/9.0 (4.0 eqv.) · 3× UVic Graduate Award	Jan 2020 – Apr 2022
Grand Valley State University — B.S. (honors), Chemistry; Biology minor; Green Chemistry cert. · GPA 3.6/4.0	Aug 2016 – Apr 2019
ACS Division of Organic Chemistry Undergraduate Award · GVSU Sustainability Champion Award	

RESEARCH & TRANSLATIONAL WORK

Northwestern University — PhD Candidate, Tyo Lab	Sep 2023 – Present
• Developing data-integrated metabolic models of the gut microbiome to predict diet- and therapeutic-driven community dynamics — feeding directly into microbiome-targeted intervention design.	
Argonne National Laboratory — Assistant Computational Biologist, Henry Lab	Jun 2021 – Present
• Scientist pursuing numerous collaborations across basic environmental sciences	
• Core developer of ModelSEED / ModelSEEDpy — an advanced open platform for genome-scale metabolic reconstruction.	
• Curator of openTECR , an open thermodynamics database underpinning rational metabolic engineering.	
Icahn School of Medicine at Mount Sinai — Bioinformatician, Karr Lab	Dec 2020 – Apr 2022
• Built BioSimulators and BioSimulations — community-standard registries and a cloud platform enabling reproducible biological simulation across the field.	
Lawrence Berkeley National Lab — Software Engineer, Reactive Transport (Molins)	Jun 2020 – Oct 2020
• Reactive-transport modeling of geochemical scaling in reverse-osmosis desalination — translated into a pre-print.	
University of Victoria — Software Engineer, Buckley Lab (Green Safe Water)	Jan 2020 – Apr 2022
• Simulations of bacterial biophysics & desalination geochemistry; first-author M.A.Sc. thesis.	
NC State & Shaw University — Metabolic Spectroscopist, Lucia Lab / GenoVerde BioSciences	May 2019 – Dec 2019
Washington State University — NSF REU Fellow, Wolcott Lab (cellulose photocatalysis)	Summer 2018
Grand Valley State University — Bioanalytical Chemist, Kovacs Lab (willow metabolomics)	Oct 2016 – Jun 2019

SELECTED PUBLICATIONS (180 CITATIONS · H-INDEX 5 · FULL LIST AT www.ANDREWFREIBURGER.COM)

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- Kennedy MS, **Freiburger A**, et al. "Diet outperforms microbial transplant to drive microbiome recovery in mice." *Nature*, 2025.
 - Borton MA, ..., **Freiburger A**, et al. "A functional microbiome catalogue crowdsourced from North American rivers." *Nature*, 2025.
 - Wood-Charlson EM, Henry CS, ..., **Freiburger A**, et al. "KBase: Open-source platform for collaborative biological data analysis." *J. Molecular Biology*, 2026.
 - Shaikh B, ..., **Freiburger AP**, et al. "BioSimulators: a central registry of simulation engines and services." *Nucleic Acids Research*, 2022.
 - Faria JP, ..., **Freiburger AP**, et al. "ModelSEED v2: High-throughput genome-scale metabolic model reconstruction." *bioRxiv*, 2023.

OPEN-SOURCE PLATFORMS & TECHNICAL SKILLS

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- **Lead developer:** *CommScores* and *CommPhitting* — Python toolkits for scoring and fitting microbial community models.
 - **Core contributor:** *ModelSEEDpy*, *BioSimulators*, *openTECR*, *Kbase*.
 - **Stack:** Python (COBRApy, pandas, scikit-learn), Jupyter, Git, Docker, HTML/CSS/JS, PHREEQC, MATLAB.

ENTREPRENEURIAL TRAINING & LEADERSHIP

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- **NSF Innovation Corps (I-Corps) Bay Area graduate** — customer-discovery and commercialization training at LBNL, 2020.
 - **Co-organizer & logistics lead**, ACS National Organic Symposium (NOS) 2025: <https://www.nationalorganicsymposium.org/>.
 - **Webmaster**, ACS Division of Geochemistry (2022–Present) · **Asst. Webmaster**, ACS Division of Organic Chemistry (2020–Present)
 - **Instructor** of 200-level Life Cycle Analysis course at the University of Victoria (2021).